

RW4 Synthetic Consumer

INTERPRETATION GUIDE

- ADF $p > 0.05$ and KPSS $p < 0.05$ -> likely $I(1)$ -like; prefer differences/cointegration models.
- ADF $p < 0.05$ and KPSS $p > 0.05$ -> stationary; level models are more admissible.
- Ljung-Box $p < 0.05$ -> autocorrelation; add lag structure.
- BP/White $p < 0.05$ -> heteroskedasticity; use robust or HAC standard errors.
- JB $p < 0.05$ -> non-normality; use robust inference and avoid strict normality assumptions.
- Stability flag=1 -> potential breaks/drift; use rolling or split-sample robustness.
- For retail transmission, compare no-promo vs promo-controlled estimates.

RUN NOTES

link_rows=3

RW4 domestic vertical chain is FarmGateUA -> ProducerUA -> ProZorro -> Retail, estimated with both forward and reverse-flow pairs. Farm-gate enters from two alternative reconstruction workbooks and both linear/pchip interpolation variants are carried end-to-end. Retail_combined is the anchored downstream index built from Silpo effective prices, Novus observed prices, and a level-aligned ConsumerUA anchor, while Retail_combined_core keeps the strict retailer-only overlap for robustness.

Table: Synthetic_Consumer_Link

product	standardized_type	n_obs	coef_retail	p_retail	coef_producer	p_producer	r2
Молоко питне	milk	56	0.0006170987828229427	0.5777992398736067	0.943225223379526	7.577082686107432e-29	0.8723404713710914
Сир твердий	hard_cheese	56	0.00020376027608469318	0.3489522548733358	0.9255256838346126	0.0	0.9791246197782543
Сметана	sour_cream	51	-0.0002863434322670274	0.8956552864195457	1.1003703971747334	3.991164062063931e-230	0.988665326159806